USER SCENARIOS

Smart Waste Management System

Scenario 1: Bin Monitoring and Overflow Detection

Scenario: A waste operations manager wants to identify bins that are close to overflow and prioritize them for collection.

Main Sequence:

1. The manager logs into the Smart Waste Management System dashboard.
2. The manager opens the Bin Monitoring module.
3. The system displays simulated bin fill levels, bin type, location, and operational status on the map.
4. The manager filters the view to show bins that have exceeded the alert threshold.
5. The system highlights full or nearly full bins using color-coded indicators.
6. The manager reviews the affected zones and selects the bins that require urgent attention.
7. The system stores the selected bins as collection priorities for route planning.

Alternative Sequence:

- If bin data is temporarily unavailable, the system displays the last recorded value and marks the bin for refresh.
- If a bin is marked as damaged, the system flags it for maintenance instead of regular collection.

Special Requirements:

- Real-time or simulated fill-level updates.
- Color-coded map visualization for bin status.
- Threshold-based alert detection.

Post-condition: The manager has an updated list of high-priority bins for operational planning.

Scenario 2: Route Optimization for Daily Collection

Scenario: A manager wants to generate an efficient waste collection route based on current bin status and truck capacity.

Main Sequence:

1. The manager accesses the Route Optimization module.
2. The system retrieves the current list of bins requiring collection.
3. The manager selects the collection zone, available truck, and driver assignment.
4. The manager enters route constraints such as truck capacity and shift duration.
5. The system calculates an optimized collection route based on priority bins, location, and operational constraints.
6. The optimized route is displayed on the map with stop order and estimated completion time.
7. The manager reviews the generated route and confirms it for execution.
8. The system assigns the route to the selected driver.

Alternative Sequence:

- If the selected truck does not have enough capacity, the system suggests a different truck or split route.
- If route optimization cannot be completed due to missing data, the system notifies the manager and allows manual adjustment.

Special Requirements:

- Fast route calculation for operational use.
- Support for truck capacity, zone boundaries, and shift constraints.
- Map-based route visualization.

Post-condition: An optimized and approved route is assigned for waste collection.

Scenario 3: Driver Executes Assigned Route

Scenario: A driver wants to complete daily collection tasks using the route assigned by the system.

Main Sequence:

1. The driver logs into the driver interface on a mobile device.
2. The system displays the assigned route, stop sequence, and list of bins to be collected.
3. The driver starts the shift and activates route navigation.
4. The driver travels to each stop and performs collection activities.
5. After servicing a bin, the driver marks the stop as completed in the system.
6. The system updates route progress and collection status in real time.
7. The driver finishes the route and submits the shift summary.

Alternative Sequence:

- If road access is blocked, the driver reports the issue and the system suggests a route adjustment.
- If internet connectivity is lost, the driver continues working offline and the data is synchronized when the connection is restored.

Special Requirements:

- Mobile-friendly interface for drivers.
- Offline support for field operations.
- Step-by-step route and stop management.

Post-condition: The assigned collection route is completed and recorded in the system.

Scenario 4: Fleet Tracking and Vehicle Status Monitoring

Scenario: A fleet manager wants to track trucks and monitor their operational status during collection activities.

Main Sequence:

1. The fleet manager opens the Fleet Management dashboard.
2. The system displays all active trucks with simulated location, status, assigned driver, and route information.
3. The fleet manager selects a specific truck to view detailed information.
4. The system shows fuel usage, maintenance status, and route progress for the selected vehicle.

5. The fleet manager compares active vehicles across zones and identifies delays or issues.
6. The system updates truck status continuously during working hours.

Alternative Sequence:

- If the vehicle location feed is interrupted, the system displays the last known location and status timestamp.
- If a truck reports a breakdown, the system alerts the manager and marks the vehicle as unavailable.

Special Requirements:

- Real-time or near real-time vehicle visibility.
- Truck assignment and operational status monitoring.
- Support for fuel and maintenance indicators.

Post-condition: The fleet manager has an updated view of vehicle operations and can intervene when needed.

Scenario 5: Vehicle Maintenance Management

Scenario: A maintenance coordinator wants to record and manage truck servicing activities.

Main Sequence:

1. The maintenance coordinator logs into the Fleet Management module.
2. The coordinator opens the maintenance records for a selected truck.
3. The system displays service history, current condition, and upcoming maintenance schedule.
4. The coordinator records a new maintenance event or updates the vehicle condition.
5. The system stores the maintenance information and recalculates the next service date.
6. If the vehicle is unavailable for operations, the system marks it as under maintenance.

Alternative Sequence:

- If a scheduled service is overdue, the system generates a maintenance alert.
- If maintenance data is incomplete, the system prompts the coordinator to provide the missing details.

Special Requirements:

- Maintenance history tracking.
- Automatic service reminders and alerts.
- Secure update of operational vehicle status.

Post-condition: Vehicle maintenance information is updated and reflected in fleet planning.

Scenario 6: Citizen Reports a Waste-Related Issue

Scenario: A citizen wants to report an overflowing bin or illegal dumping through the platform.

Main Sequence:

1. The citizen opens the Citizen Reporting interface.
2. The citizen enters the issue location using map selection or device location.
3. The citizen selects the issue type and writes a short description.
4. The citizen uploads a photo of the issue.
5. The system validates the submitted information.
6. The citizen submits the report.
7. The system creates a new report entry and assigns it a tracking status.
8. The system notifies the responsible manager or field operator.

Alternative Sequence:

- If the photo upload fails, the system allows the citizen to retry or submit the report without the image.
- If the location is incomplete, the system prompts the citizen to provide a valid location before submission.

Special Requirements:

- Simple reporting workflow for non-technical users.
- Secure handling of citizen data and uploaded media.
- Automatic issue registration and assignment.

Post-condition: The citizen's report is successfully recorded and forwarded for action.

Scenario 7: Manager Assigns and Tracks Citizen Reports

Scenario: A manager wants to review incoming citizen reports and assign them to the appropriate field staff.

Main Sequence:

1. The manager logs into the Citizen Reporting module.
2. The system displays a list of new and active reports with location, priority, and status.
3. The manager opens a selected report to review its details and attached photo.
4. The manager assigns the report to a field operator or links it to an active collection route.
5. The system updates the report status to indicate that it is in progress.
6. After the issue is resolved, the manager or operator updates the report status to resolved.
7. The system stores the full status history of the report.

Alternative Sequence:

- If the report is duplicated, the system allows the manager to merge it with an existing case.
- If no operator is available, the system keeps the report pending and highlights it for follow-up.

Special Requirements:

- Status tracking throughout the report lifecycle.
- Role-based assignment and update permissions.
- Audit trail for issue handling.

Post-condition: Citizen complaints are assigned, tracked, and resolved through the system.

Scenario 8: Recycling Monitoring and Contamination Control

Scenario: A recycling supervisor wants to monitor recycling bins and detect contamination issues.

Main Sequence:

1. The supervisor opens the Recycling Management module.
2. The system displays recycling bin locations, fill levels, and contamination alerts.
3. The supervisor reviews bins flagged as contaminated or improperly used.
4. The system shows recent collection records and contamination history for the selected recycling bin.
5. The supervisor decides whether to schedule collection, inspection, or awareness action.

6. The system records the supervisor's decision and updates the recycling status.

Alternative Sequence:

- If contamination data is unavailable, the system marks the bin for manual inspection.
- If contamination exceeds the acceptable level, the system generates a high-priority alert.

Special Requirements:

- Separate handling of recycling operations from general waste.
- Support for contamination alerts and recycling performance tracking.
- Historical record of recycling bin activity.

Post-condition: Recycling bins are monitored properly and contamination issues are identified for action.

Scenario 9: Reporting and Analytics for Operational Decisions

Scenario: An operations manager wants to generate reports and review KPIs for waste collection performance.

Main Sequence:

1. The manager opens the Reporting and Analytics module.
2. The system displays available report types and KPI dashboards.
3. The manager selects a report period, zone, or operational category.
4. The system generates charts, summaries, and performance indicators such as collected bins, response time, truck utilization, and recycling rate.
5. The manager reviews the dashboard and compares trends across zones.
6. The manager exports the report in PDF or CSV format for sharing or record-keeping.

Alternative Sequence:

- If the selected dataset is incomplete, the system notifies the manager and marks missing values clearly.
- If export fails, the system allows the user to retry the export process.

Special Requirements:

- Accurate KPI calculation and trend visualization.
- Support for PDF and CSV export.
- Near real-time analytics refresh.

Post-condition: The manager receives a clear operational report for decision-making and municipal reporting.

Scenario 10: System Handles Dynamic Route Changes During Operations

Scenario: The system needs to react when new urgent bins or incidents appear while collection is already in progress.

Main Sequence:

1. The system continuously monitors active routes, new alerts, and incoming citizen reports.
2. A new urgent event is detected, such as an overflowing bin or a high-priority complaint.
3. The system evaluates the location of the active trucks and the urgency of the new task.
4. The system recommends a route adjustment for the most suitable driver or team.
5. The manager reviews the suggested route change and approves it.
6. The updated route is sent to the driver's mobile interface.
7. The driver receives the change and continues collection using the revised route.

Alternative Sequence:

- If the manager rejects the proposed change, the original route remains active and the new task is queued separately.
- If no available truck can handle the urgent task, the system escalates the issue and marks it for manual scheduling.

Special Requirements:

- Continuous event monitoring during operations.
- Dynamic route recalculation based on operational priorities.
- Immediate communication of route updates to field staff.

Post-condition: Urgent operational events are integrated into live collection activities with minimal disruption.